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1  /*
2  * EECE.3220: Data Structures
3  * Instructor: M. Geiger
4  * Class design example
5  *
6  * orchardTest.cpp: main program
7  */
8
9  #include "Orchard.h"
10 #include "Apple.h"
11
12 #include <fstream>
13 #include <iostream>
14
15 using namespace std;
16
17 int main() {
18     string fname;        // Input file name
19     ifstream inFile;     // Input file stream
20     string cmd;          // Input command
21
22     Orchard oList[10];   // List of orchards
23     unsigned nOrch = 0;  // Number of orchards
24
25     string oName;        // Orchard name
26     unsigned op, cl;     // Open/close time
27     unsigned ntypes;     // Number of types
28
29     string aName;        // Apple name
30     unsigned m, ntrees;  // Month available; number of trees
31
32     unsigned i;          // Loop index
33
34     do {
35         cout << "Enter command: ";
36         cin >> cmd;
37
38         // Read new orchard information from file
39         if (cmd == "read") {
40             cout << "Enter input file name: ";
41             cin >> fname;
42             inFile.open(fname);
43
44             // Input file format: orchard name on 1st line, open/close times on 2nd, # types on 3rd
45             // Each apple type on 2 lines: type name on first, month & numberOfTrees on 2nd
46             getline(inFile, oName);
47             inFile >> op >> cl >> ntypes;
```

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48
49     /**** WRITE INPUT INFO INTO Orchard OBJECT IN ARRAY ****/
50
51     for (i = 0; i < ntypes; i++) {
52         inFile.ignore(1);          // Skip newline at end of      ↗
53         previous line
54         getline(inFile, aName);
55         inFile >> m >> ntrees;
56
57         /**** WRITE INPUT INFO INTO Orchard OBJECT IN ARRAY ****/
58     }
59
60     inFile.close();
61
62     nOrch++;          // Count new orchard
63 }
64
65 // Print all information about given orchard
66 else if (cmd == "print") {
67     if (nOrch == 0)
68         cout << "No current orchards listed\n";
69
70     else if (nOrch == 1)
71         ;    /***** DISPLAY STATE OF CURRENT ORCHARD *****/
72
73     else {
74         cout << "Enter orchard number between 0 and " << nOrch - 1 ↗
75         << '\n';
76         cin >> i;
77         ;    /***** DISPLAY STATE OF CURRENT ORCHARD *****/
78     }
79 }
80
81 // Check availability of apple types
82 else if (cmd == "available") {
83     if (nOrch == 0)
84         cout << "No current orchards listed\n";
85
86     else {
87         cout << "Enter month: ";
88         cin >> m;
89
90         if (nOrch == 1)
91             ;    /***** DISPLAY LIST OF AVAILABLE APPLES IN ONLY  ↗
92             ORCHARD *****/
93
94         else {
95             cout << "Enter orchard number between 0 and " << nOrch ↗
```

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- 1 << '\n';
94     cin >> i;
95     ;    /***** DISPLAY LIST OF AVAILABLE APPLES IN GIVEN ORCHARD *****/
96         }
97     }
98 }
99
100     else if (cmd != "exit")
101         cout << "Error: invalid command " << cmd << '\n';
102
103 } while (cmd != "exit");
104
105 return 0;
106 }
```